Latchan Chhetri

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Research Profile

I am a researcher in the final year of a B.Tech in CSE (AI & Data Science), with first-author & senior-author works at IEEE, ACM, Springer, T&F and CVF-track venues. My research has two connected threads. The first is designing fast, compact vision models: convolutions and hybrid CNN/state-space backbones that stay accurate under tight compute budgets. The second is checking whether reported gains actually hold up, using different splits, out-of-distribution and cross-domain tests, conformal coverage and interpretability, so that a model's limits are visible rather than hidden behind a single benchmark number. I test these ideas across medical imaging, remote sensing, space environments, underwater restoration and damaged-text recognition.

Education

Manipal Institute of Technology (Sikkim), Sikkim Manipal University — Sikkim, India
B.Tech, Computer Science & Engineering (Artificial Intelligence & Data Science) — *Jul 2023 – Jun 2027*

St. Xavier's School — West Bengal, India
ISC (Class XII), Science — *2023*

Publications

** equal contribution* † *corresponding author*

Conference Papers (Accepted / Published)

[1] A. Kumar*, **L. Chhetri***†, D. Das*. "SPECTRAFORGE: Domain-Equalized Frequency-Spatial Fusion for Synthetic Dermatology Detection." ***Accepted for Short Oral Presentation,*** *IEEE International Conference on Data Science and Advanced Analytics (DSAA), 2026.* **[CORE A Tier | <25% Acceptance Rate]**

[2] **L. Chhetri**†, A. Kumar, D. Das, P. Ghosal. "Interpretable Solar Panel Defect Detection via Fuzzy Rule Extraction from Hierarchical Vision Models." *Indian Conference on Computer Vision, Graphics and Image Processing (ICVGIP), 2026. Accepted, ACM ICPS.*

[3] **L. Chhetri**†, A. Kumar. "DDV-GNet: High-Throughput Defect Detection for Space Manufacturing via Deep Delta Gated Networks." *IEEE SPACE, 2026.* ***Oral presentation.***

[4] **L. Chhetri**†, A. Datta, P. Ghosal. "Attention-Enhanced Swin Transformers for Robust Brain Tumor Classification Under Patient-Level Data Splitting." *IEEE GCON, 2026.* doi:10.1109/GCON69192.2026.11648823

[5] **L. Chhetri**†, A. Kumar. "Optimizing Deep Learning for Brain Tumor Classification: A Comparative Ablation Study of Preprocessing and Augmentation Strategies." *IEEE GCON, 2026.* doi:10.1109/GCON69192.2026.11649062

[6] A. Mukherjee, A. Kumar, S. Bala, S. R. Varma, A. Anand, H. Das, **L. Chhetri**†. "Risk-Controlled Urban Change Detection: Conformal Prediction Wrappers for Provable Reliability in High-Resolution Satellite Imagery." *ICCI, 2026.* ***Oral presentation.*** (senior and corresponding author)

[7] R. Das†, **L. Chhetri**, A. Kumar, P. Ghosal. "Intrinsic Neural Firewalls for Cyber-Physical Systems: Robust Anomaly Rejection via Deep Delta Residual Overwrites." *WIN 6.0, 2026.* ***Oral presentation.***

Preprints

[8] **L. Chhetri***†, A. Anand*. "Slice, Subtract, Segment: Efficient 2D and Physiologically Informed 3D Brain Tumor Segmentation." *bioRxiv, 2026.* doi:10.64898/2026.10.04.756528

[9] A. Kumar*, A. Anand*, **L. Chhetri***†. "DVARA: Frequency Gates for Adapting Frozen Underwater Image Enhancement Networks." *Preprints.org, 2026.* doi:10.20944/preprints202610.1474.v1

[10] A. Kumar*, A. Anand*, **L. Chhetri***†. "Trust, but Verify: Epistemic Uncertainty-Gated Pseudo-Labeling for Safer Semi-Supervised Malaria Screening." *bioRxiv, 2026.* doi:10.64898/2026.10.13.758712

Submitted & Under Review

[11] **L. Chhetri**†, A. Anand. "JALDRISHTI: Seeing Text Through Water with Fold-Aware, Physics-Gated Diffusion." *IEEE/CVF CVPR 2027.*

[12] **L. Chhetri**†, A. Kumar, D. Das, P. Ghosal. "Point Two Ways at Once: Decoupled Erase-Write Refinement for Convolutional and State-Space Vision Backbones." *IEEE/CVF CVPR 2027.*

[13] **L. Chhetri**†, D. Das, A. Kumar. "Don't Add, Route: Resolving Operator Basis Mismatch in Hybrid Medical Segmenters via Spectral Delta Fusion." *IEEE/CVF CVPR 2027.*

[14] A. Kumar, **L. Chhetri**†, D. Das, P. Ghosal. "Learn the Delta, Not the Storm: Physics-Informed Residual Gating for Cross-Cycle TEC Generalization." *IEEE/CVF CVPR 2027.*

Research Experience

Research Intern, CVPR Unit — Jun 2026 – Present
Indian Statistical Institute, Kolkata (jointly with the University of Salford, UK) — *Hybrid*
Supervisors: Prof. Umapada Pal (ISI Kolkata) and Prof. Shivakumara Palaiahnakote (University of Salford)

- Working on recognition of licence plates that are bent, buckled, torn or partly covered, where standard OCR models fail badly on real damaged images. The core question is when the visible character evidence is enough to correct the geometry, and when the information is truly lost.
- Developing a recognition-guided feedback loop that corrects local geometry only when the remaining character evidence supports it, and returns candidates with uncertainty when it does not. Designed experiments to separate deformation errors from missing-information errors before building the full model.

- Benchmarked multiple recognizers (PARSeq, LPRNet, TrOCR, PaddleOCR, Qwen2.5-VL, etc) on a manually verified set of crash-damaged plates. The VLM had the lowest character error rate (0.49) but read zero plates fully correctly: it stopped at folds and substituted plausible-looking characters for damaged ones.
- Building a Blender-based synthetic generator to supply dense deformation, damage and visibility ground truth, and collected a manually verified real damaged-plate test set.

Research Collaborator, Remote Sensing & EO Vision Sep 2026 – Present
Prof. Swalpa Kumar Roy (Tezpur University) and Dr. Saurabh Kaushik (University of Wisconsin–Madison) *Hybrid*
- Adapting pretrained Earth-observation foundation models (e.g., THOR) to downstream tasks, modifying architectures and fine-tuning strategy for semantic segmentation and localization in satellite imagery.
- Setting up a multi-dataset evaluation across optical satellite benchmarks to test how well foundation-model features transfer between sensors, resolutions and regions, rather than tuning to a single benchmark.
- Reproducing recently published remote sensing models from their released code to establish trustworthy baselines, after finding that several reported results could not be reproduced.

Research Assistant, Satellite AI & Causal Inference Jun 2026 – Present
AI & Global Development Lab (AIDevLab), UT Austin / Chalmers University *Remote*
Supervisors: Prof. Connor T. Jerzak (UT Austin) and Prof. Adel Daoud (Chalmers University)
- Studying how conformal prediction intervals behave when a satellite-imagery poverty model is moved to a new region; showing that standard intervals lose their coverage under large geographic shift.
- Wrote the data loaders for continuous International Wealth Index (IWI) regression and set up baselines and coverage metrics for geographic hold-outs, for example training on African data and testing outside the continent.

Founder & Lead Researcher Jan 2026 – Present
Halo Mind Research Group *Onsite*
- Started and run a small group of undergraduate researchers working on efficient vision architectures and reliable evaluation. Group work includes the DSAA 2026 and ICCI 2026 papers and the bioRxiv brain tumor segmentation preprint.
- Set research directions, plan experiments, review code, and write papers with junior members. As senior author on the ICCI paper, I supervised a seven-person team from idea to oral presentation. Currently preparing a CVPR 2027 submission.

Research Assistant & Team Lead Jul 2025 – Jul 2026
Sikkim Manipal Institute of Technology *Onsite*
- Showed that image-level train/test splits inflate brain tumor classification accuracy by up to 3.71%. Proposed a Swin Transformer with CBAM attention that reached 96.82% accuracy under patient-level splitting and had the smallest leakage gap of five models (IEEE GCON 2026).
- Ran controlled preprocessing experiments on 3,064 MRI scans from 233 patients. Found that skull stripping and CLAHE hurt accuracy while bilateral filtering with augmentation reached 93.86% (AUC 0.991) (IEEE GCON 2026).
- Engineered Intrinsic Neural Firewalls utilizing Deep Delta Residual Overwrites for edge-deployed cyber-physical systems. Achieved high-performance, zero-shot anomaly rejection against False Data Injection Attacks (WIN 6.0).
- Directed and mentored student research teams, training junior researchers in core deep learning methodologies and guiding end-to-end experimental design from conceptualization to multiple first author and co-authored acceptances in IEEE, Springer, T&F venues.

AI & Data Science Intern Jul 2025 – Aug 2025
Soft Nexis Technology *Remote*
- Engineered and deployed **RealVisor**, an end-to-end AI real estate platform. Developed robust predictive pipelines utilizing XGBoost and Random Forest Regressors to estimate property valuations based on complex spatial features.
- Conducted rigorous evaluation across curated real-world datasets, strictly outperforming baseline models, and designed a production-ready Streamlit dashboard featuring interactive market trend visualizations and automated investment analysis.

Ongoing Research Projects

AHF-RBFNet: Fuzzy Attention for Medical Image Segmentation Ongoing
With Dr. Swalpa Kumar Roy and Dr. Palash Ghosal | Target: Q1 journal
- Replaced the Gaussian fuzzy membership in a hierarchical-fusion U-Net with a learnable RBF kernel, applied at the bottleneck and on all four skip connections to sharpen uncertain lesion boundaries.
- ISIC 2016: 84.55% IoU and 91.97% Dice, ahead of every baseline in the original AHF-U-Net comparison, including AHF-U-Net itself. Extending to ISIC 2017 and BUSI, with zero-shot transfer to PH2 and a full ablation of each RBF component.

Degraded Ashokan Brahmi Script Recognition Ongoing
CV lead, with A. Anand, A. Kumar | Target: ICDAR 2027
- Generated 20K+ synthetic Brahmi characters with WGAN-GP and applied physically motivated degradations to build a 150K-sequence training set.
- Recognition model: SimCLR-pretrained ResNet-34 encoder with a BiLSTM-CTC decoder. Building a severity-graded character error rate (CER) benchmark to measure the synthetic-to-real gap for ancient Indic scripts.

Patent

L. Chhetri et al. "Halo-Hydra: A Physics-Informed Generative Framework for Multimodal Visual Restoration in Underwater, Turbid and Degraded Scattering Environments." Indian Patent Office / PCT, filed (pending), 2026. A lightweight generative model, designed for edge deployment, that restores underwater and hazy images in real time without needing clean reference images for training.

Academic Service & Honors

- **Technical Peer Reviewer (ICVGIP 2026):** Nominated and invited by Program Chairs to evaluate research manuscripts for the **Indian Conference on Computer Vision, Graphics and Image Processing** (Published by **ACM ICPS** / supported by **IUPRAI**).
- **Technical Peer Reviewer (IEEE GCON 2026):** Invited official peer reviewer evaluating submissions in applied deep learning, medical imaging, and computer vision architectures for **IEEE GCON**.
- **NPTEL Elite + Gold, Topper (Top 1%):** Introduction to Internet of Things, IIT Kharagpur. Scored 91%; top 1% of 50,282 certified candidates.
- **Online Coursework:** Deep Learning with PyTorch for Medical Image Analysis (Udemy); Introduction to Applied Machine Learning (Amii, via Coursera); Natural Language Processing with Deep Learning in Python (Udemy); Data Science & AI Masters (Udemy).

Technical Skills

Languages	Python, C, Java, LaTeX
Deep Learning	PyTorch, TensorFlow/Keras, Optuna
Architectures	CNNs, Vision Transformers, State-Space Models (Mamba's), U-Net variants, GANs, Diffusion Models, VLMs, Foundation Models
Methods	Segmentation, Detection, Classification, OCR, Self-Supervised Learning, Conformal Prediction, OOD Evaluation, Explainable AI, Interpretability
Data & Tools	NumPy, Pandas, SciPy, Scikit-Learn, Blender, ETL Workflows, Hadoop, Spark, Spacy, NLTK, Git, Streamlit